# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 9 — MATHS · MEDIUM LEVEL MCQ TEST SET 1

**Full Syllabus (Chapters 1–14)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

## MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

*Choose the correct option and write it in your answer book.*

**1.** The square of the distance between $(1, 4)$ and $(-3, 2)$ is: 1

**(a)** 17 **(b)** 25 **(c)** 20 **(d)** 22

**2.** If $x = 2$ is a zero of $7x + m$, then $m =$ 1

**(a)** -15 **(b)** -14 **(c)** -13 **(d)** -12

**3.** Which of the following has a terminating decimal expansion? 1

**(a)** $\frac{39}{65}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{14}{21}$ **(d)** $\frac{5}{18}$

**4.** $(7x + 4y)^2 =$ 1

**(a)** $49x^2 - 56xy + 16y^2$ **(b)** $49x^2 + 28xy + 16y^2$ **(c)** $49x^2 + 16y^2$ **(d)** $49x^2 + 56xy + 16y^2$

**5.** The length of a chord of a circle of radius $5$ cm lying $4$ cm from the centre is (in cm): 1

**(a)** 7 **(b)** 5 **(c)** 8 **(d)** 6

**6.** The length of the arc of a sector of angle $90°$ in a circle of radius $14$ cm (take $\pi = \frac{22}{7}$) is (cm): 1

**(a)** 21 **(b)** 24 **(c)** 22 **(d)** 23

**7.** Two dice are rolled together. The probability that the sum is 7 is: 1

**(a)** $\frac{1}{6}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{7}$

**8.** In an AP, $t_1 = 7$ and $t_{11} = 47$. The common difference is: 1

**(a)** $4$ **(b)** $5$ **(c)** $6$ **(d)** $3$

**9.** The converse of "If a number is divisible by 6, then it is divisible by 3" is false because: 1

**(a)** $x = -3$ gives $x^2 = 9$ but $x \neq 3$

**(b)** a kite that is not a rhombus has perpendicular diagonals

**(c)** $9$ is divisible by 3 but not by 6

**(d)** two angles of $60°$ in an equilateral triangle are equal but not vertically opposite

**10.** The average of 12 numbers is $89$. When a number $63$ is included, the average of all 13 numbers becomes: 1

**(a)** $87$ **(b)** $89$ **(c)** $86$ **(d)** $88$

**11.** Using Euclid's subtraction algorithm for gcd$(360, 156)$, the number of subtraction steps needed is: 1

**(a)** $8$ **(b)** $11$ **(c)** $10$ **(d)** $9$

**12.** The angles of a quadrilateral are in the ratio $5:3:3:1$. The largest angle is: 1

**(a)** $170^\circ$ **(b)** $140^\circ$ **(c)** $150^\circ$ **(d)** $160^\circ$

**13.** The slope of the line $2x + 2y = 10$ is: 1

**(a)** $0$ **(b)** $-2$ **(c)** $-1$ **(d)** $1$

**14.** The radius of a sphere is multiplied by $6$. Its surface area becomes ___ times: 1

**(a)** 216 **(b)** 12 **(c)** 6 **(d)** 36

**15.** The midpoint of $(3, 2)$ and $(6, 7)$ is: 1

**(a)** $(5,\ 4.5)$ **(b)** $(4.5,\ 4.5)$ **(c)** $(-1.5,\ -2.5)$ **(d)** $(4.5,\ 5.5)$

**16.** The slope of the line through $(1, 3)$ and $(3, 9)$ is: 1

**(a)** 3 **(b)** 5 **(c)** 4 **(d)** 2

**17.** The number of decimal places in the decimal expansion of $\frac{7}{400}$ is: 1

**(a)** 6 **(b)** 4 **(c)** 3 **(d)** 5

**18.** $(2x + 3)(3x + 5) =$ 1

**(a)** $6x^2 + x + 15$ **(b)** $6x^2 + 19x + 15$ **(c)** $6x^2 + 20x + 15$ **(d)** $6x^2 + 19x - 15$

**19.** A chord of length $8$ cm lies $3$ cm from the centre of a circle. The radius (in cm) is: 1

**(a)** 6 **(b)** 4 **(c)** 7 **(d)** 5

**20.** A triangle has area $84$ cm$^2$ and one side $14$ cm. The altitude on this side (cm) is: 1

**(a)** 12 **(b)** 14 **(c)** 13 **(d)** 11

**21.** Two dice are rolled together. The probability of getting a double is: 1

**(a)** $\frac{1}{6}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{6}$

**22.** In an AP, $d = 5$ and $t_{13} = 69$. The first term is: 1

**(a)** $8$ **(b)** $11$ **(c)** $10$ **(d)** $9$

**23.** Which of the following is a counterexample to the statement “Every multiple of 3 is odd”? 1

**(a)** $3$ **(b)** $6$ **(c)** $9$ **(d)** $15$

**24.** The average of 27 numbers is $84$. If the number $58$ is removed, the average of the remaining 26 numbers is: 1

**(a)** $86$ **(b)** $85$ **(c)** $87$ **(d)** $84$

**25.** Using Euclid's subtraction algorithm for gcd$(420, 364)$, the number of subtraction steps needed is: 1

**(a)** $8$ **(b)** $9$ **(c)** $11$ **(d)** $10$

**26.** One angle of a parallelogram exceeds another adjacent angle by $70^\circ$. The smaller angle is: 1

**(a)** $90^\circ$ **(b)** $65^\circ$ **(c)** $55^\circ$ **(d)** $50^\circ$

**27.** The points $(3, 6)$, $(5, 10)$ and $(7, k)$ are collinear. The value of $k$ is: 1

**(a)** $14$ **(b)** $13$ **(c)** $15$ **(d)** $16$

**28.** The edge of a cube is multiplied by $5$. Its volume becomes ___ times: 1

**(a)** 125 **(b)** 15 **(c)** 5 **(d)** 25

**29.** One end of a diameter of a circle is $(5, 1)$ and the centre is $(2, -1)$. The other end is: 1

**(a)** $(1, -3)$ **(b)** $(-1, 3)$ **(c)** $(-3, -1)$ **(d)** $(-1, -3)$

**30.** The $x$-intercept of the line $y = 4x + 12$ is: 1

**(a)** -2 **(b)** -3 **(c)** -4 **(d)** -1

**31.** $0.\overline{12}$ in lowest terms is: 1

**(a)** $\frac{4}{33}$ **(b)** $\frac{5}{33}$ **(c)** $\frac{2}{17}$ **(d)** $\frac{1}{11}$

**32.** If $a - b = 3$ and $ab = 18$, then $a^2 + b^2 =$ 1

**(a)** 47 **(b)** 45 **(c)** 46 **(d)** 44

**33.** In a circle with centre $O$, chord $AB$ has $\angle AOB = 64^\circ$. Then $\angle OBA =$ 1

**(a)** $48^\circ$ **(b)** $78^\circ$ **(c)** $58^\circ$ **(d)** $68^\circ$

**34.** The height of a parallelogram of area $204$ cm$^2$ and base $17$ cm is (cm): 1

**(a)** 12 **(b)** 13 **(c)** 14 **(d)** 11

**35.** A die is rolled. The probability of getting a number less than 3 or a number greater than 4 is: 1

**(a)** $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{4}$

**36.** The 4th and 9th terms of an AP are $15$ and $35$. Its 15th term is: 1

**(a)** $61$ **(b)** $60$ **(c)** $59$ **(d)** $58$

**37.** For the expression $n^2 + n + 101$, the smallest natural number $n$ for which the value is NOT prime is: 1

**(a)** $5$ **(b)** $4$ **(c)** $6$ **(d)** $3$

**38.** The average of 40 numbers is $73$. The average of all 60 numbers (including 20 more numbers) is $70$. The average of the 20 additional numbers is: 1

**(a)** $64$ **(b)** $63$ **(c)** $65$ **(d)** $66$

**39.** A counter starts at 0. For i = 1, 2, …, 50, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

**(a)** $25$ **(b)** $22$ **(c)** $24$ **(d)** $23$

**40.** The diagonal of a rectangle is $34$ cm and one side is $16$ cm. Its area is: 1

**(a)** $490$ cm$^2$ **(b)** $510$ cm$^2$ **(c)** $480$ cm$^2$ **(d)** $496$ cm$^2$

**41.** The line $4x + 2y = 9$ is parallel to the line $kx + 4y = 11$. The value of $k$ is: 1

**(a)** $10$ **(b)** $7$ **(c)** $9$ **(d)** $8$

**42.** The radii of two spheres are in the ratio $1 : 6$. The ratio of their volumes is: 1

**(a)** $1 : 6$ **(b)** $1 : 210$ **(c)** $1 : 36$ **(d)** $1 : 216$

**43.** The point on the $x$-axis equidistant from $(1, 5)$ and $(5, -5)$ is: 1

**(a)** $(4, 0)$ **(b)** $(-3, 0)$ **(c)** $(3, 0)$ **(d)** $(0, 3)$

**44.** The point $(k, 31)$ lies on $y = 7x + 3$. Then $k =$ 1

**(a)** 6 **(b)** 4 **(c)** 3 **(d)** 5

**45.** The length of the repeating block in the decimal expansion of $\frac{1}{7}$ is: 1

**(a)** 8 **(b)** 5 **(c)** 6 **(d)** 7

**46.** If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2} =$ 1

**(a)** 9 **(b)** 8 **(c)** 6 **(d)** 7

**47.** The angles of a cyclic quadrilateral $ABCD$ at $A$ and $C$ are in the ratio $2 : 3$. Then $\angle A =$ 1

**(a)** $62^\circ$ **(b)** $92^\circ$ **(c)** $72^\circ$ **(d)** $82^\circ$

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**48.** A square has area $169$ cm$^2$. Its perimeter (cm) is: 1

**(a)** $54$ **(b)** $51$ **(c)** $52$ **(d)** $53$

**49.** A bag has 3 red and 5 blue balls. After adding 4 more red balls, the probability of drawing a red ball is: 1

**(a)** $\frac{2}{3}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{7}{13}$ **(d)** $\frac{7}{12}$

**50.** The sum of the first 13 even natural numbers is: 1

**(a)** $195$ **(b)** $182$ **(c)** $208$ **(d)** $169$

**END OF QUESTION PAPER** **Total: 50 Marks**

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